

See, Sense, Spark: Profiling Granularity in Proactive AI for Creative Research Ideation

Gahui Kim

Sungkyunkwan University
Seoul, Republic of Korea
Immersive media engineering
Seoul, Republic of Korea
gahee020907@g.skku.edu

Yebom Choi

Sungkyunkwan University
Seoul, Republic of Korea
Department of Human-Artificial
Intelligence Interaction
Seoul, Republic of Korea
yebom618@g.skku.edu

Changjun Lee

Sungkyunkwan University
Seoul, Republic of Korea
School of Convergence
Seoul, Republic of Korea
changjunlee@skku.edu

Abstract

Proactive AI systems are a new paradigm for supporting cognitively demanding tasks by anticipating user needs and providing timely interventions. While prior creativity research often suggests that broad, less constrained information cues can enhance associative flexibility, it remains unclear how deeply these systems should understand users to improve their thinking. This paper explores the psychological effects of profiling granularity on creative research ideation. Using a proactive AI system called *See, Sense, Spark*, we conducted a between-subjects study comparing coarse and fine-grained user profiling. Contrary to our initial hypothesis, participants with fine-grained profiles showed significantly higher I-type curiosity. This result establishes profiling granularity as a core design parameter for proactive AI, highlighting that topical relevance can outweigh breadth in stimulating exploratory reasoning. We conclude by suggesting the need for adaptive systems that can balance both breadth and relevance to foster meaningful human-AI collaboration in complex cognitive tasks.

ACM Reference Format:

Gahui Kim, Yebom Choi, and Changjun Lee. 2025. See, Sense, Spark: Profiling Granularity in Proactive AI for Creative Research Ideation. In *SIGGRAPH Asia 2025 Posters (SA Posters '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3757374.3771458>

1 Introduction

Proactive artificial intelligence (AI) systems are reshaping human-computer interaction by anticipating user needs and delivering timely interventions. While higher personalization is often assumed to increase engagement, creativity research suggests a more nuanced link between informational breadth and creative output. Divergent thinking [Guilford 1967] and analogical transfer theories [Gick and Holyoak 1980] indicate that broad, less constrained cues can enhance associative flexibility and foster novel ideas. Likewise, studies in design ideation [Brown 2009] show that diverse, loosely

connected stimuli promote creative exploration. From this perspective, coarse-grained profiling, offering broad, less targeted prompts, may facilitate serendipitous connections.

In contrast, fine-grained profiling can narrow the ideational space and, as noted in the automation trust literature [Hoff and Bashir 2015], make system errors more salient, potentially reducing engagement. We therefore hypothesized that coarse-grained profiling would better support curiosity and epistemic agency, defined as the capacity to actively shape knowledge goals, strategies, and evaluation, than fine-grained profiling. We examine six factors: perceived usefulness, satisfaction, trust, epistemic agency, and two types of curiosity—interest-driven (I-type) and deprivation-driven (D-type) [Litman 2008].

2 Methodology

2.1 System Overview

The study proposed a proactive AI system called *See, Sense, Spark* to support creative research ideation. Implemented on *XReal Air 2 Pro* smart glasses, the system captured users' visual surroundings and academic context in real time, rendering micro-prompts on the glasses' display. To prevent users from being overwhelmed, the system prioritized prompts that were both stimulating and contextually relevant. Each AI-generated prompt was evaluated using the following formula: $Score = Novelty \times (Major Alignment + Delight + Utility + Exploration Potential)$

All components were normalized to either 0 or 1; $Score \geq 2.0$ indicated sufficient combined value to warrant display. Only prompts meeting this threshold were refined and shown on the smart glasses, thereby minimizing information overload [Cai et al. 2025].



Figure 1: Lab Experiment and User View

2.2 Experimental Design

A between-subjects experiment was conducted to examine how profile granularity affects psychological responses. Participants were randomly assigned to one of two conditions. The prompt rate and

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA Posters '25, Hong Kong, Hong Kong

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2134-2/25/12

<https://doi.org/10.1145/3757374.3771458>

length were held constant across conditions, and a manipulation check was performed using lexical diversity (Type-Token Ratio on fixed-length segments) and domain-term incidence.

Group A (Coarse profile): The AI in this group had access only to basic user data, including age, gender, occupation, and academic discipline ($n = 10$).

Group B (Fine-grained profile): The AI in this group had access to all the coarse profile data plus a detailed academic profile, which included information on participants' research stage, key research topics, recent scholarly focus, and relevant journals ($n = 10$).

Each session followed a structured protocol: (1) a 5-minute consent process and pre-survey, (2) 15 minutes of system interaction in a simulated research setting, and (3) a 10-minute survey and interview. Interviews were conducted by a conversational AI on the ElevenLabs using pre-scripted prompts to standardize delivery.

2.3 Participants

The study included a total of twenty researchers with a mean age of 26.4. Three participants were excluded from the final analysis due to extreme composite scores on preregistered attention/compliance checks ($|z| > 2.5$), resulting in a final sample size of 17 (Group A: $n = 8$; Group B: $n = 9$). All procedures were approved by the IRB.

3 Results

Lexical analysis using TTR on fixed-length segments confirmed that profiling manipulation was effective, showing higher diversity in outputs for the fine-grained group (0.90 vs. 0.86, $p < .001$). The fine-grained group output included more domain-specific terms such as 'agent' and 'simulation,' whereas the coarse-grained group output used more generic terms such as 'image' and 'desk.'

3.1 Quantitative outcomes

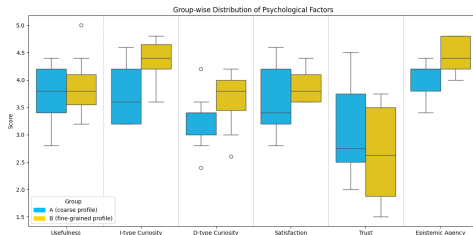


Figure 2: Factor Scores by Group

We compared six psychological constructs based on profile granularity. The fine-grained profile group (Group B) demonstrated significantly higher I-type curiosity ($p = .016$, $d = 1.30$) and a non-significant trend toward higher epistemic agency ($p = .066$, $d = 0.98$). This suggests that fine-grained profiling enhances interest-driven engagement and perceived authorship.

K-means clustering identified four user clusters: two that aligned with the experimental conditions (Cluster 1: 4B; Cluster 2: 5A) and two that exhibited emergent patterns. Cluster 0, labeled "Goal-Oriented Pragmatists," exhibited high usefulness and low D-type curiosity. In contrast, Cluster 3, labeled "Open Explorers," showed

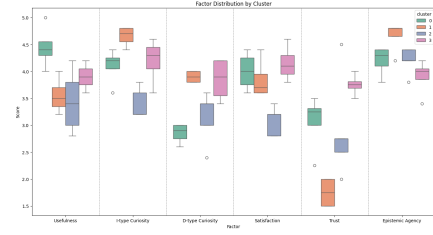


Figure 3: Factor Distribution by Cluster

high D-type curiosity and satisfaction. ANOVA indicated a significant difference in D-type curiosity ($p = .002$), which was identified as a primary predictor for cluster assignment through decision tree analysis. These findings suggest that proactive systems can create a need to fill knowledge gaps, while user-driven, reflective engagement fosters deeper exploration based on personal interest.

3.2 Qualitative Theme

Following the experiment, participants were interviewed by an AI interviewer to minimize interviewer effects, ensuring consistent clarity and depth in their verbal responses. Two researchers independently coded the responses into four categories aligned with the quantitative constructs.

Fine-profile users more frequently exhibited *Contextual Curiosity* and *Affective Engagement*, corresponding to their higher I-type curiosity and satisfaction scores. By contrast, coarse-profile users emphasized *Pragmatic Utility*, consistent with their higher usefulness scores. *Reflective Reframing* occurred at similar rates in both groups, aligning with the absence of a significant difference in D-type curiosity.

Table 1: Qualitative–Quantitative Alignment by Theme

Theme	A(%)	B(%)	Construct	Quantitative Result
Reflective Reframing	16	15	D-type Curiosity	No difference
Contextual Curiosity	20	36	I-type Curiosity	$B > A$ ($d = 1.30$)
Affective Engagement	13	18	Satisfaction	$B > A$ ($d \approx 0.7$)
Pragmatic Utility	51	31	Perceived Usefulness	$A > B$ ($d = 0.34$)

Findings contradicted our hypothesis. Fine-grained profiling resulted in significantly higher I-type curiosity and showed a trend toward higher epistemic agency. Increased topical relevance in fine-grained prompts may have provided stronger cognitive anchors, linking new ideas to prior knowledge; relevance can outweigh breadth in stimulating curiosity and higher-order thinking.

4 Conclusion

This study demonstrates that profiling granularity in proactive AI meaningfully shapes users' psychological experiences during creative research ideation. Contrary to our hypothesis favoring coarse breadth, fine-grained profiling produced greater interest-driven curiosity and affective engagement, with qualitative evidence of richer contextual and emotional responses, whereas coarse profiles emphasized pragmatic utility.

Treating granularity as a core design parameter, future work should build adaptive systems balancing breadth and relevance, specify when coarse profiling is advantageous (e.g., early-stage ideation), and test generalizability beyond the present context.

Acknowledgments

This research was supported by the MSIT(Ministry of Science and ICT), Korea, under the Graduate School of Metaverse Convergence support program(IITP-2025-RS-2023-00254129) supervised by the IITP(Institute for Information & Communications Technology Planning & Evaluation)

References

- Tim Brown. 2009. *Change by design: How design thinking creates new alternatives for business and society*. Harper Business, New York, NY, USA.
- Runze Cai, Nuwan Janaka, Hyeongcheol Kim, Yang Chen, Shengdong Zhao and Yun Huang, and David Hsu. 2025. AiGet: Transforming Everyday Moments into Hidden Knowledge Discovery with AI Assistance on Smart Glasses. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 631, 26 pages. doi:10.1145/3706598.3713953
- Mary L. Gick and Keith J. Holyoak. 1980. Analogical problem solving. *Cognitive Psychology* 12, 3 (1980), 306–355. doi:10.1016/0010-0285(80)90013-4
- J.P. Guilford. 1967. *The Nature of Human Intelligence*. McGraw-Hill, New York, NY, USA.
- Kevin Hoff and Masooda Bashir. 2015. Trust in automation: Integrating empirical evidence on factors that influence trust. *Human Factors* 57, 3 (2015), 407–434. doi:10.1177/0018720814547570
- Jordan A. Litman. 2008. Interest and deprivation factors of epistemic curiosity. *Personality and Individual Differences* 44, 7 (2008), 1585–1595. doi:10.1016/j.paid.2008.01.014